

L^2 -acyclicity and cost for a rank-100 amalgam

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Abstract

We give a direct calculation of all L^2 -Betti numbers of the rank-100 amalgam in OpenAI’s fixed-price construction. An elementary regular-tree argument makes its cellular boundary injective, proves its displayed presentation complex aspherical, and gives $\beta_n^{(2)}(\Gamma) = 0$ for every $n \geq 0$, independently of either cost bound. Combining this calculation with OpenAI’s positive Bernoulli cost estimate gives an explicit free ergodic probability-measure-preserving orbit relation with cost at least $1 + (100 \cdot 2^{61})^{-1}$ and vanishing L^2 -Betti numbers. The group infimum cost remains one. The construction and positive cost estimate are due to OpenAI; the cost–Betti comparison and action invariance are due to Gaboriau. This is an attributed consequence and verification note, with a separate direct invariant proof.

1 Question, conventions, and attribution

For a countable probability-measure-preserving (p.m.p.) equivalence relation $\mathcal{R}$ on a standard atomless probability space, the cost is the infimum, over generating graphings, of the sum of the domain measures of their partial isomorphisms. Gaboriau proved

$$\text{Cost}(\mathcal{R}) - 1 \geq \beta_1^{(2)}(\mathcal{R}) - \beta_0^{(2)}(\mathcal{R}) \quad (1)$$

and asked whether equality always holds [2, Corollary 3.23 and p. 129]; see also his FAQ [3, p. 1]. The zeroth term is essential. Infinite classes have $\beta_0^{(2)} = 0$; a free action of a finite group of order q instead has $\beta_0^{(2)} = 1/q$, $\beta_1^{(2)} = 0$, and cost $1 - 1/q$.

For clarity, we write

$$\text{Cost}_*(\Gamma) = \inf\{\text{Cost}(\mathcal{R}_\alpha) : \alpha \text{ is an essentially free p.m.p. action of } \Gamma\}$$

for group infimum cost [1, Definition I.5]. The action-specific equality, the relation-level equality, and equality for this infimum are distinct assertions; Gaboriau’s notes state them separately [4, Question 2.65].

The group and both cost estimates used below are OpenAI’s [5, Theorem 1.1]. Its manuscript is dated October 5, 2026; the public repository version checked here is the October 6 release at commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. The general implication from a failure of fixed price to a cost–Betti counterexample was already explicit in Popa–Shlyakhtenko–Vaes [6, Remark 6.6], whose first arXiv version appeared in 2018. The cost-limit deduction for this particular group was already recorded conditionally in our public preliminary triage [9, §7 and accompanying audit]. This note expands that observation with an independent cellular calculation of all L^2 -Betti numbers and a focused validation record for the upstream cost input. No independent fixed-price discovery or priority for the classical implication is asserted. The bounded priority audit found no earlier instance of the particular direct calculation below; that search does not certify novelty.

2 Explicit group and actions

Let

$$A = F(a, b_1, \dots, b_{99}), \quad w = ab_1ab_2 \cdots ab_{99}a, \quad J = \langle b_1, \dots, b_{99}, w \rangle \leq A,$$

and put $B = J \times \langle t \rangle$, where $\langle t \rangle \cong \mathbb{Z}$. Define

$$\Gamma = A *_J B = \langle a, b_1, \dots, b_{99}, t \mid [t, b_i] = 1 \ (1 \leq i \leq 99), [t, w] = 1 \rangle. \quad (2)$$

We use right actions. On $X_0 = [0, 1]^\Gamma$ with product Lebesgue probability, the Bernoulli action is $(xg)(h) = x(gh)$. Let X be its invariant conull set of points with trivial stabilizer, with restricted probability μ .

Lemma 1. *The group Γ is countably infinite and generated by the displayed 101 elements. There is a surjection $\chi : \Gamma \rightarrow \mathbb{Z}$ with $\chi(a) = 1$ and $\chi(b_i) = \chi(t) = 0$. For every positive integer M , the action on*

$$Y_M = X \times \mathbb{Z}/M\mathbb{Z}, \quad (x, r)g = (xg, r + \chi(g) \bmod M),$$

with product probability and uniform height coordinate, is free, ergodic, and p.m.p., as is the action on X .

Proof. Amalgam normal form embeds A and B in Γ . In particular Γ is infinite and t has infinite order. The displayed generators generate both factors. The generators of J are free: every nonempty reduced word in the b_i and w expands to a nonempty reduced word in A , since w starts and ends with a , w^{-1} with a^{-1} , and opposite adjacent w -symbols are excluded by reducedness. This also gives the presentation in (2).

The exponent sum of a on A agrees on J with the map $jt^n \mapsto \chi_A(j)$ on B , so it extends to χ . Notice $\chi(w) = 100$. Coordinate permutations preserve product measure. For $g \neq 1$, $xg = x$ implies $x(g) = x(1)$, a null event for independent continuous coordinates. Countability gives the invariant conull free set X . The formula for Y_M satisfies the action law and preserves product measure; freeness follows from its first coordinate.

The transformation $x \mapsto xt$ is mixing. Indeed, two cylinder events become independent under t^n for all sufficiently large n : their finite coordinate supports become disjoint because t has infinite order. Approximation in measure by cylinders proves mixing for measurable events. A Γ -invariant subset of X is therefore null or conull. For a Γ -invariant measurable subset of Y_M , each height section is t -invariant and hence null or conull. Translation by a cycles the heights and equates their measures. Thus the subset is null or conull. The countable and finite products used here are standard atomless probability spaces. $\square$

3 Direct cellular computation

Put $u_0 = 1$, $u_i = u_{i-1}ab_i$ for $1 \leq i \leq 99$, and $S = \sum_{i=0}^{99} u_i$. Recall the free-basis substitution used in [5, §5.2]: $a, u_1, \dots, u_{99}$ are a free basis of A , with inverse substitutions $b_i = a^{-1}u_{i-1}^{-1}u_i$. The substitutions in both directions recover the generators by induction. Thus $F = \langle u_1, \dots, u_{99} \rangle$ is free of rank 99 and embeds in Γ .

Lemma 2. *Right convolution by S and by $t - 1$ has trivial kernel on $\ell^2(\Gamma)$.*

Proof. Form a bipartite graph with two copies F_L, F_R of F and edges $g_L - (gu_i)_R$, $0 \leq i \leq 99$. It is connected and 100-regular. A nonbacktracking cycle would give $u_{i_1}u_{j_1}^{-1} \cdots u_{i_k}u_{j_k}^{-1} = 1$ with consecutive edge labels distinct. Delete the factors $u_0 = 1$. No two zeros are adjacent; deletion

joins factors of the same sign, and every remaining opposite-sign adjacent pair has distinct indices. A nonidentity factor remains, yielding a nonempty freely reduced word. Hence the graph is the infinite 100-regular tree.

Adjacency on an infinite d -regular tree, $d \geq 2$, has no ℓ^2 kernel at zero. Root the tree at o and put $E_n = \sum_{\text{dist}(o,v)=n} |f(v)|^2$. If adjacency kills f , at each vertex of depth $n \geq 1$ the sum on its $d-1$ children is minus its parent's value. Cauchy–Schwarz and summation over depth $n \geq 2$ give $E_{n+1} \geq E_{n-1}$, since every parent is counted $d-1$ times. The odd and positive even layer sums are nonnegative, nondecreasing, and summable, hence zero. The equation at a neighbor of o then gives $f(o) = 0$.

For $(Tf)(g) = \sum_i f(gu_i)$, adjacency is $\begin{pmatrix} 0 & T \\ T^* & 0 \end{pmatrix}$, so both T and T^* are injective. Right convolution by S on coefficient functions is T^* : $(fS)(g) = \sum_i f(gu_i^{-1})$. Decomposition over left cosets gF transfers injectivity to $\ell^2(\Gamma)$. A vector killed by right multiplication by $t-1$ is constant on every infinite right- t orbit and thus zero by square summability. The operators are bounded finite sums of translations. $\square$

Proposition 3. *The presentation complex in (2) is a finite $K(\Gamma, 1)$ of dimension two. For every $n \geq 0$, $\beta_n^{(2)}(\Gamma) = 0$.*

Proof. Choose $r_i = tb_i t^{-1} b_i^{-1}$ and $r_w = twt^{-1} w^{-1}$. Write coefficients on the left of chosen lifted cells. The completed cellular boundaries are left- Γ -equivariant and use right multiplication in coordinates:

$$\ell^2(\Gamma)^{100} \xrightarrow{d_2} \ell^2(\Gamma)^{101} \xrightarrow{d_1} \ell^2(\Gamma), \quad d_1(f) = \sum_x f_x(x-1).$$

The lifted attaching path contributes its prefix before each positive letter and minus its prefix after each negative letter, equivalently the classical Fox derivative [7]. Put $P_i = u_{i-1}a$. Then $D_a w = S$ and $D_{b_i} w = P_i$. Evaluating in Γ gives, for $q = (q_1, \dots, q_{99}, q_w)$,

$$\begin{aligned} (d_2 q)_a &= q_w(t-1)S, \\ (d_2 q)_{b_i} &= q_i(t-1) + q_w(t-1)P_i, \\ (d_2 q)_t &= \sum_{i=1}^{99} q_i(1-b_i) + q_w(1-w). \end{aligned}$$

The factor order is fixed; none is commuted past S or P_i . The identity $S(a-1) + \sum_i P_i(b_i-1) = w-1$ also checks $d_1 d_2 = 0$. If $d_2 q = 0$, its a -coordinate and Lemma 2 give $q_w(t-1) = 0$, then $q_w = 0$. The b_i -coordinates give $q_i = 0$ for every i . Thus d_2 is injective.

This restricts to injectivity on finitely supported integral cellular chains. The universal cover is simply connected and two-dimensional, so its first homology and homology above degree two vanish; injectivity kills its second homology. A simply connected acyclic CW complex is contractible by Hurewicz and Whitehead. The presentation complex is therefore a $K(\Gamma, 1)$.

The orthogonal complement of $\text{im } d_1$ consists of vectors invariant under all generators, hence constant on the infinite group and zero in ℓ^2 . Its image is dense. Von Neumann dimension additivity for finite Hilbert modules [2, Properties 1.2(1),(3), p. 106], or [8, Theorem 1.11 and §2.3], gives

$$\dim_\Gamma \ker d_1 = 101 - 1 = 100, \quad \dim_\Gamma \overline{\text{im } d_2} = 100.$$

The quotient has dimension zero, so $\beta_1^{(2)} = 0$. Dense image gives $\beta_0^{(2)} = 0$, injectivity gives $\beta_2^{(2)} = 0$, and there are no higher cells. The Euler characteristic $1 - 101 + 100 = 0$ is consistent with the result; by itself it would only give $\beta_1^{(2)} = \beta_2^{(2)}$. $\square$

4 Cost input and exact parameter

For

$$K = e^{96} \frac{96^{99}}{95^{95}}, \quad 0 < \alpha < \frac{1}{200}, \quad K\alpha^3 < \frac{1}{2},$$

OpenAI's theorem gives

$$\text{Cost}(\mathcal{R}_X) \geq 1 + \frac{\alpha}{100}, \quad \text{Cost}(\mathcal{R}_{Y_M}) \leq 1 + \frac{99}{M} \quad (M > 100). \quad (3)$$

Here $\mathcal{R}_Z$ denotes the orbit relation of Γ on Z . The upper estimate has a short direct proof, retained to make the route to vanishing checkable independently of the positive cost input.

Proposition 4. *The second bound in (3) holds for every integer $M > 100$.*

Proof. Fix $0 < p < 1$. The Borel set

$$C_p = (\{x \in X : x(1) < p\} \cup \{x \in X : x(t^n) \geq p \text{ for all } n \in \mathbb{Z}\}) \times \mathbb{Z}/M\mathbb{Z}$$

has measure p and meets every t -orbit. The second set is null by independence of the distinct t^n coordinates and corrects the exceptional orbits. Keep the full t -map and the 100 generators b_i, w on C_p . These generate the B -relation: for any such j and any z , choose n with $zt^n \in C_p$, and use $z \rightarrow zt^n \rightarrow zt^n j \rightarrow zj$, since j commutes with t . Their cost is $1 + 100p$.

Add the a -map from heights $0, \dots, 98$, costing $99/M$. Put $u_0 = 1$, $u_i = u_{i-1}ab_i$, so $w = u_{99}a$ and $\chi(u_i) = i$. If every a -edge from heights $k, \dots, k + 98$ is available, take z at height $k + 99$ and $y = zu_{99}^{-1}$ at height k . The path $y \rightarrow yu_1 \rightarrow \dots \rightarrow yu_{99} = z$ uses those a -edges and the supplied b_i -edges. The supplied w -edge also joins y to $yw = za$. It provides the next a -edge. Starting with the initial 99 heights and taking $k = 0, \dots, M - 100$ provides all remaining source heights through $M - 1$, including the final edge to height zero. The graphing therefore generates the Γ -relation. Its cost is $1 + 100p + 99/M$ for every $p > 0$; taking the infimum proves the bound. $\square$

We fix $\alpha = 2^{-61}$. Its admissibility is certified without floating-point arithmetic:

$$K = e^{96} 96^4 (1 + 1/95)^{95} < e^{97} 96^4 < 3^{97} 96^4, \quad 2 \cdot 3^{97} 96^4 < 2^{183}. \quad (4)$$

The last inequality is an exact integer comparison, reproduced in the supplement; $e < 3$ and $(1 + 1/95)^{95} < e$ give the preceding bounds. Also $2^{61} > 200$. Consequently $K\alpha^3 < 1/2$ and $0 < \alpha < 1/200$. The positive gap supplied by the first bound in (3) is

$$\eta = \frac{1}{100 \cdot 2^{61}}.$$

5 The invariant calculation

Theorem 5. *For the group (2) and its free Bernoulli orbit relation $\mathcal{R}_X$,*

$$\beta_n^{(2)}(\Gamma) = \beta_n^{(2)}(\mathcal{R}_X) = 0 \quad (n \geq 0), \quad \text{Cost}(\mathcal{R}_X) \geq 1 + \frac{1}{100 \cdot 2^{61}} > 1.$$

More generally every essentially free p.m.p. action of Γ has every L^2 -Betti number zero. Its group infimum cost satisfies $\text{Cost}_(\Gamma) = 1$. In particular $\mathcal{R}_X$ has strict inequality in (1), while $\text{Cost}_*(\Gamma) = 1 + \beta_1^{(2)}(\Gamma)$.*

Proof. Proposition 3 and Gaboriau’s action-invariance theorem [2, Corollary 3.16, p. 126] give $\beta_n^{(2)}(\mathcal{R}_Z) = \beta_n^{(2)}(\Gamma) = 0$ for every $n \geq 0$ and every essentially free p.m.p. action Z . This theorem does not impose ergodicity. Restricting to an invariant conull free set does not change the relation or its measured invariants. Every orbit is infinite, since Γ is infinite. Thus $\beta_0^{(2)}(\mathcal{R}_Z) = 0$ by [2, Properties 3.15(1), p. 126].

As a second proof of first-Betti vanishing, already recorded in the preliminary triage, apply (1) to every Y_M and use Proposition 4. The same group invariant satisfies

$$0 \leq \beta_1^{(2)}(\Gamma) = \beta_1^{(2)}(\mathcal{R}_{Y_M}) \leq \text{Cost}(\mathcal{R}_{Y_M}) - 1 \leq \frac{99}{M} \quad (M > 100).$$

For every $\epsilon > 0$, choose an integer $M > \max\{100, 99/\epsilon\}$. Hence $\beta_1^{(2)}(\Gamma) = 0$. Action invariance gives the asserted vanishing for X and every other free action, and gives $\beta_0^{(2)}(\Gamma) = 0$. The first bound in (3), with (4), yields the asserted positive cost gap.

For every free action, (1) now gives cost at least one. The sequence Y_M gives infimum at most $1 + 99/M$ for every $M > 100$. Thus $\text{Cost}_*(\Gamma) = 1$. This infimum assertion does not assert that some action attains cost one. $\square$

6 Dependency and verification scope

The positive cost bound remains the substantial input from OpenAI. Its proof proceeds through relative graphing compression and deployment, transfer to ranks of finite permutation presentations, and expansion with a coefficient-sensitive planar shortening argument. The supplementary verification record identifies the exact statements, source hashes, independent proof audits, and boundary checks for those inputs. The proof of Theorem 5 uses that input explicitly; it does not independently establish failure of fixed price. The direct all-degree vanishing proof in Proposition 3 uses neither cost bound. The group infimum-cost calculation additionally uses Proposition 4. Both remain valid independently of the Bernoulli lower-bound argument.

The source’s `INPUTS.md` says that its intermediate source record does not establish that it was the final revision. The version actually used is pinned and the current-public-version check is recorded in the supplement. No relevant Lean formalization was identified in the source catalogue, and no result here is claimed machine-formalized. The package provides proof-audit artifacts and reproducible exact arithmetic rather than a formal proof certificate of the upstream construction.

Disclosure. AI tools were used extensively in research, drafting, source checking, and adversarial verification. Automated reviews are documented in the supplement. At publication this preprint has not undergone conventional human peer review or refereeing. The cited upstream manuscript is attributed to OpenAI according to its supplied citation. No affiliation or coauthor is asserted.

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